

# Nikhil Reddy Billa

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## SUMMARY

Applied scientist working on Agentic AI, research in Responsible AI, LLM safety, adversarial robustness, and evaluation for agentic systems. Publications at ICLR 2026 and EMNLP 2025, plus 2+ years of production ML engineering.

## EXPERIENCE

### SAP Labs

Jun 2026 – Present

*Applied AI Scientist – Agentic AI*

*Palo Alto, CA*

- Develop synthetic data-generation pipelines for agentic query-translation models, targeting underrepresented failure scenarios to broaden fine-tuning coverage.
- Built an evaluation framework for natural-language-to-query models emitting OData/REST queries, measuring retrieval correctness across S/4HANA, Ariba, and SuccessFactors.
- Extend ToolSense, a diagnostic benchmark auditing how well LLMs encode parametric tool knowledge, targeting tool-selection and argument-grounding failure modes across agentic workflows.

### Responsible Data Science (ReDS) Lab, Virginia Tech

Oct 2024 – May 2026

*Graduate Research Assistant - Advisor: Dr. Ruoxi Jia*

*Blacksburg, VA*

- Adversarial Deja Vu (ICLR 2026, with Princeton and Amazon AGI), compressed 16,901 adversarial skills mined from 32 jailbreak papers into 397 dictionary primitives via sparse dictionary learning, training on skill compositions cut ASR on post-cutoff unseen attacks.
- Confusion-Inducing Attack (EMNLP Main 2025, with Cisco Research), quantified training-data leakage by targeting spikes in token-level prediction entropy, achieved a 22.2% verbatim extraction rate on Llama-2 70B.
- Developed an end-to-end evaluation and finetuning pipeline for the Amazon Nova Trusted AI Challenge using LoRA/QLoRA and GRPO to optimize the Pareto frontier between code utility and security.
- Diagnosed multi-agent failures on GAIA and SWE-bench by backtracking OpenTelemetry execution traces to localize and categorize the causal failure span in long agentic trajectories.
- Automated deployment of jailbreak vectors (PAIR, AutoDAN-Turbo, Crescendo-MT) into a unified red-teaming CI/CD pipeline, enabling iterative adversarial hardening between training runs.
- Built and containerized 13 LLM-as-a-Judge evaluation harnesses integrating CodeQL, Amazon CodeGuru, and Semgrep for high-throughput security assessment of generated code (Amazon Nova AI Challenge).

### NCR Voyix

Jul 2022 – Aug 2024

*Software Engineer, Machine Learning (MLOps)*

*Hyderabad, India*

- Built a LayoutLM-based document classifier over an OCR-noisy internal corpus of 450 scanned documents across 9 classes, lifting accuracy 34% over the incumbent rule-based extractor by fine-tuning on layout and token embeddings jointly.
- Cut median time-to-detection on latency regressions from ~50 min to ~4 min across globally distributed CI/CD pipelines by clustering multi-modal log spans and surfacing the first anomalous span instead of manual triage.
- Deployed an anomaly-detection engine over system logs using transformer-based sequence modeling; flagged 20+ P2-severity incidents pre-impact over 8 months, validated against the on-call incident log.
- Orchestrated MLOps infrastructure on Amazon EKS with automated CI/CD for retraining and model versioning, moving retrain-to-deploy turnaround from days to hours.

## SELECTED PUBLICATIONS

- Adversarial Deja Vu: Jailbreak Dictionary Learning for Stronger Generalization. *ICLR 2026*.
- Retracing the Past: LLMs Emit Training Data When They Get Lost. *EMNLP Main 2025*.
- Data is All You Need (Almost): Iterative Synthetic Instruction Tuning for Secure Code Generation. *Amazon Nova AI Challenge 2025*.
- IDDAW: A Benchmark for Safe and Robust Segmentation of Drive Scenes in Unstructured Traffic and Adverse Weather. *WACV 2024*.
- CNN based Image Resizing Forensics for Double Compressed JPEG Images. *Journal of Information Security and Applications*

## SKILLS

**Languages:** Python, C++ , SQL, Bash

**LLM Architectures and Post-Training:** Transformers, Attention, Tokenization, Embeddings, Fine-tuning (LoRA/QLoRA), SFT, RLHF, GRPO, Quantization, Synthetic Data Generation

**Agentic Systems and Retrieval:** Agentic AI, Tool Use and Function Calling, LangGraph, MCP, RAG, Vector Search (FAISS, Pinecone), vLLM

**Evaluation and Safety:** LLM Evaluation, Benchmark Design, LLM-as-a-Judge, Red Teaming, Adversarial Robustness, AI Safety, Model Validation, Failure-Mode Analysis

**Machine Learning:** PyTorch, Hugging Face, Deep Learning, NLP, Reinforcement Learning, Classification, Regression, Optimization

**Research and Analysis:** Experimental Design, Hypothesis Testing, A/B Testing, Error Analysis, Pandas, NumPy, Matplotlib, MLflow, W & B

**Infrastructure:** Docker, Kubernetes, AWS (EC2, EKS, S3, Bedrock), Azure, CI/CD, Git

## PROJECTS

### Agentic RAG Code Assistant with MCP Integration

Python, LangGraph, FAISS, vLLM, Filesystem/GitHub MCP

- Built an agentic assistant answering natural-language questions over local codebases, GitHub repositories, and commit history via a FAISS-backed store of chunk-level source embeddings.
- Raised context recall and answer faithfulness on a held-out question set by moving from fixed-size to AST-aware chunking and re-selecting the embedding model.

## AWARDS AND LEADERSHIP

- **Amazon Nova AI Challenge 2025** - Team HokieTokie, code generation track.
- **Technical Lead, ML4E Club**, Virginia Tech - led ML initiatives and mentored students on research projects.

## EDUCATION

### Virginia Tech

Aug 2024 – May 2026

*M.S. in Computer Engineering - GPA 3.67/4.0*

*Blacksburg, VA*

### National Institute of Technology, Rourkela

Jul 2018 – Jun 2022

*B.Tech in Electrical Engineering*

*Rourkela, India*